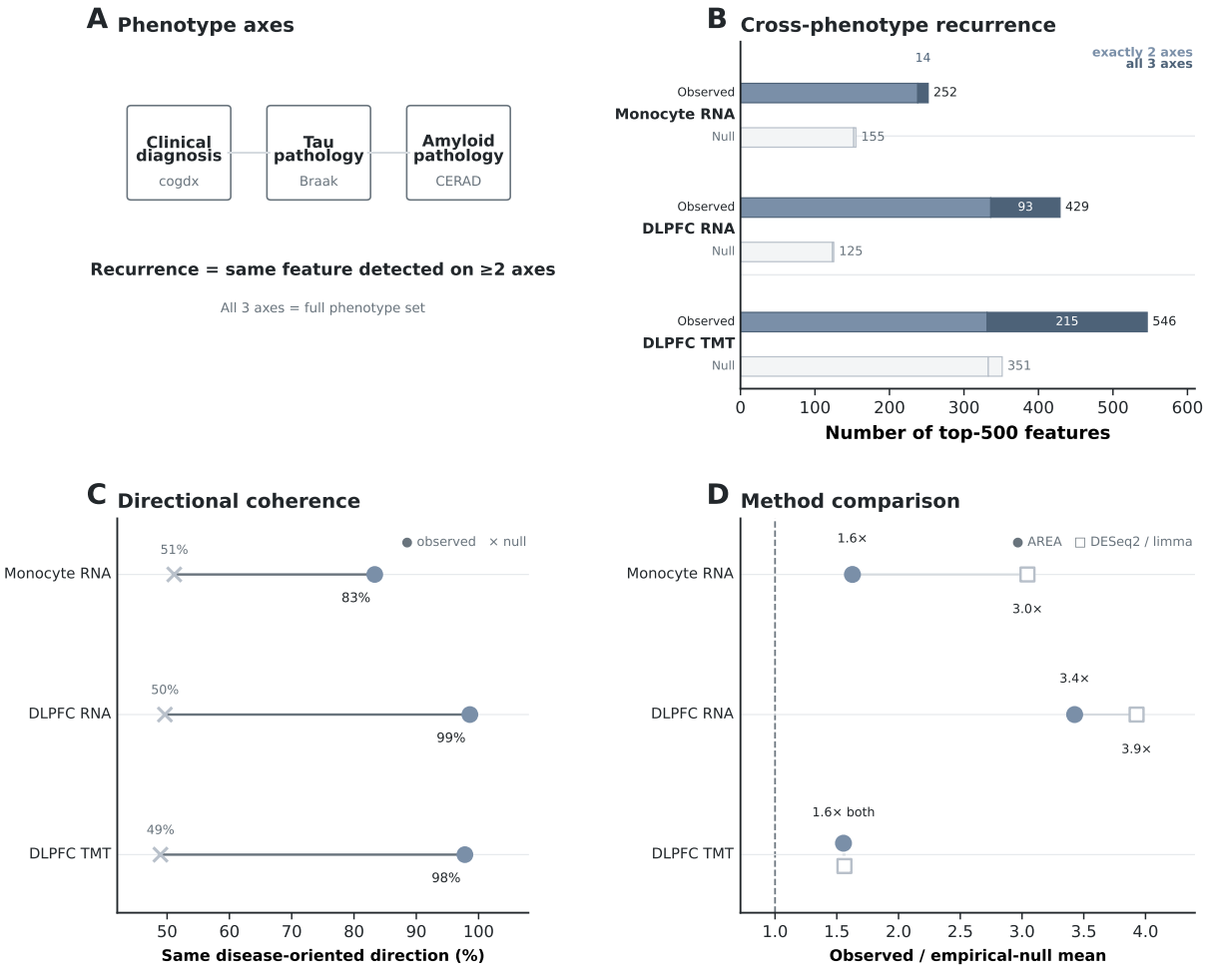
A Phenotype axes
Clinical diagnosis
cogdx
Tau pathology
Braak
Amyloid pathology
CERAD
Recurrence = same feature detected on ≥2 axes
All 3 axes = full phenotype set
B Cross-phenotype recurrence
exactly 2 axes
all 3 axes
Observed
Monocyte RNA
Null
14
252
155
Observed
DLPFC RNA
Null
93
429
125
Observed
DLPFC TMT
Null
215
546
351
0
100
200
300
400
500
600
Number of top-500 features
C Directional coherence
observed
null
Monocyte RNA
51%
83%
DLPFC RNA
50%
99%
DLPFC TMT
49%
98%
50
60
70
80
90
100
Same disease-oriented direction (%)
D Method comparison
AREA
DESeq2 / limma
Monocyte RNA
1.6×
3.0×
DLPFC RNA
3.4×
3.9×
DLPFC TMT
1.6× both
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Observed / empirical-null mean